LUMYVUE lesion detectors: how the boxes were made

Detailed documentation of the hand-boxing, the helper tools, the rules, the checks and the results (acne and hyperpigmentation)

Compiled 1 November 2026 from the labeling logs, the folders on disk, the scripts and the earlier results records. All counts below were read from the actual files (labels, `labeling_log.csv`, manifests). Nothing here is estimated unless it says so.

What this document answers

1. What a bounding box is and why the detectors need them.
2. Exactly who did what: what you did by hand, and what the AI assistant (Claude) built, drafted, checked and ran.
3. The box tool, the boxing rules, and every boxing session by folder.
4. Pictures of boxes you drew by hand, and pictures that show how the helper drafts were corrected.
5. How we measured whether the boxes are reproducible, and what that did to the scores.
6. The mistakes we made along the way.

> **Note on the pictures.** The photos come from public sources (public Roboflow datasets and web photos you collected). A few show the watermark of the original website, and some are augmented copies (stretched or rotated borders) that come from those public datasets. They are used here only to show the boxes. If this document is submitted or published, check the licence or permission for each photo, or replace them with your own consented photos. Participant photos from the kiosk are never used here.

1. What the boxes are, in plain words

A **bounding box** is a rectangle drawn around one lesion. The detector (YOLOv8) is shown thousands of photos together with the rectangles a person drew, and it learns to draw similar rectangles on new photos. So the boxes are the "answer key" for training, for choosing settings, and for the final test. If the answer key is inconsistent, the detector cannot score well, however good the model is.

Each box is saved as one line in a text file next to the photo (YOLO format): `class x_center y_center width height`, all as fractions of the photo size. A photo with no lesion has an empty text file; in the tool you press **X** to mark it.

Two kinds of boxing were needed:

- **Acne:** one small tight box per spot (papule, pustule, comedo, nodule). A face can have 5 to 30 of them. Spots are tiny: the median box is about 5 to 7 % of the photo side.
- **Hyperpigmentation:** one larger box per patch of darker skin (melasma, sun spots, post-inflammatory marks). Usually 1 to 5 boxes per photo, and the median box is about 22 to 28 % of the photo side.

2. Who did what (the honest split)

What you did by hand

- You drew or corrected **every box that is used as a label**: the first 927 acne photos from scratch, the photos reviewed after a helper draft, the hand-boxed test and validation photos, the independent and fresh acne sets, and all hyperpigmentation validation, test and training corrections under rule v2.
- You decided what counts as a lesion (with the written rules below), pressed **S** to save or **X** to mark "nothing to box" on each photo.
- Every save was timed by the tool, so the logs show about **12 hours of acne boxing and about 5.7 hours of hyperpigmentation boxing** in the listed folders (the first 646-photo hyperpigmentation round is additional); details in sections 4 and 5.
- You re-boxed photos blind to measure how well you agree with yourself, and fixed photos when the checks found problems (46 acne photos with doubled boxes, 60 acne photos with missed spots, 26 + 42 hyperpigmentation photos).

What the AI assistant (Claude) did

Job	What I did	What it means for the labels
Built the box tool (`lesion_box_tool.py`)	A local browser editor with zoom, pan, shift-click to stamp the same size, contrast view, undo, live checks for too small / too large / overlapping boxes, a review sheet, auto-save, and a time log. Nothing is uploaded; it runs on your computer.	Faster, more consistent drawing. The tool never creates boxes by itself.
Wrote the rules shown in the tool and in the documents	The acne rule, the first hyperpigmentation rule, and (with the measurements below) **annotation rule v2**.	You applied them; the rules were tested for reproducibility.
Chose and prepared photos	De-duplicated the photo pool (thumbnail correlation, 8 flips and rotations), reserved test photos, shuffled and renamed the blind sets, built the review folders (copies only, originals untouched).	Prevents the same photo from being in training and test.
Built the draft helper (`acne_predraft.py`, `hyperpig_predraft.py`)	A YOLOv8 model trained **only on your own boxes** pre-drew boxes on new photos so you only had to delete wrong ones and add missed ones.	Drafts are not labels until you review and save the photo. Unreviewed drafts were never used.
Audits and checks	Annotation audit, doubled-box check, missed-spot "second look", size and drift checks, reliability tests, the rule-v2 code conversion.	They found problems that you then fixed by hand.
Trained and scored the models	Training runs, validation-only selection, locked one-time test scoring.	Does not change any label.
Backups and progress snapshots	Copied labels and logs to a backup folder.	Safety.

What I did **not** do

- I did not draw the final labels. Machine-drafted boxes only count after you reviewed the photo in the tool and saved it.
- Early in the acne work I produced a folder of machine-drafted boxes, `claude_boxed_Acne`. **It was never reviewed in the tool and was never used**: the dataset builder explicitly skips it.
- I did not look at the test labels to tune anything. Test sets were scored once.

3. The box tool

`shared/lesion_box_tool.py` (Python standard library only; started by a `.bat` file in each boxing folder; open in the browser, port 8800).

Feature	Why it exists
Zoom (mouse wheel), pan (space + drag)	tiny acne spots are easy to see
Shift + click = stamp a box the same size as your last box	one click per acne spot, consistent size
E contrast view, **H** hide boxes	see the skin under the box (display only, the photo never changes)
Live checks: too small, too large, overlapping, unusual size compared with your other boxes (red dashed)	catches slips while you draw
V review sheet: thumbnails of finished photos with their boxes	spot mistakes at a glance
S / Enter save and next unreviewed, **X** nothing visible (saves empty), **A / D** previous / next, **N** next unreviewed	fast flow; a photo counts as **reviewed only after you save it**
Ctrl+Z / Ctrl+Y, Tab, arrow keys (Shift = resize), Delete	editing
Automatic save when you move on; every action is timed in `labeling_log.csv`	provenance and honest effort numbers

The rule for the condition is shown inside the tool (press **G**). The log row looks like: `timestamp, annotator, image, action, n_boxes, seconds`.

The acne rule (shown in the tool)

> One TIGHT box per spot (papule, pustule, comedo, nodule): the whole raised or red spot, not the skin around it. Touching spots get separate boxes when you can tell them apart. Do not box scars, freckles, moles or dark marks. No visible acne: press X.

The longer written protocol (from the annotation audit) adds: box the lesion plus a thin margin; boxes may overlap but not by more than about 30 % IoU; a cluster box only when single lesions cannot be told apart even when zoomed; do not box scars, flat marks, freckles, moles, hair or shadows; skip anything under about 8 px; in dense areas go region by region.

The first hyperpigmentation rule (shown in the tool, later replaced)

> One box per PATCH of darker skin (melasma, sun spots, post-acne marks): edge to edge around the patch, not a big region of the face. Separate patches = separate boxes. No visible patch: press X.

It did not say what a "patch" is, how far apart two patches must be, or how small a mark may be. Section 5.2 shows what that cost.

Hyperpigmentation annotation rule v2 (written 12 Oct 2026)

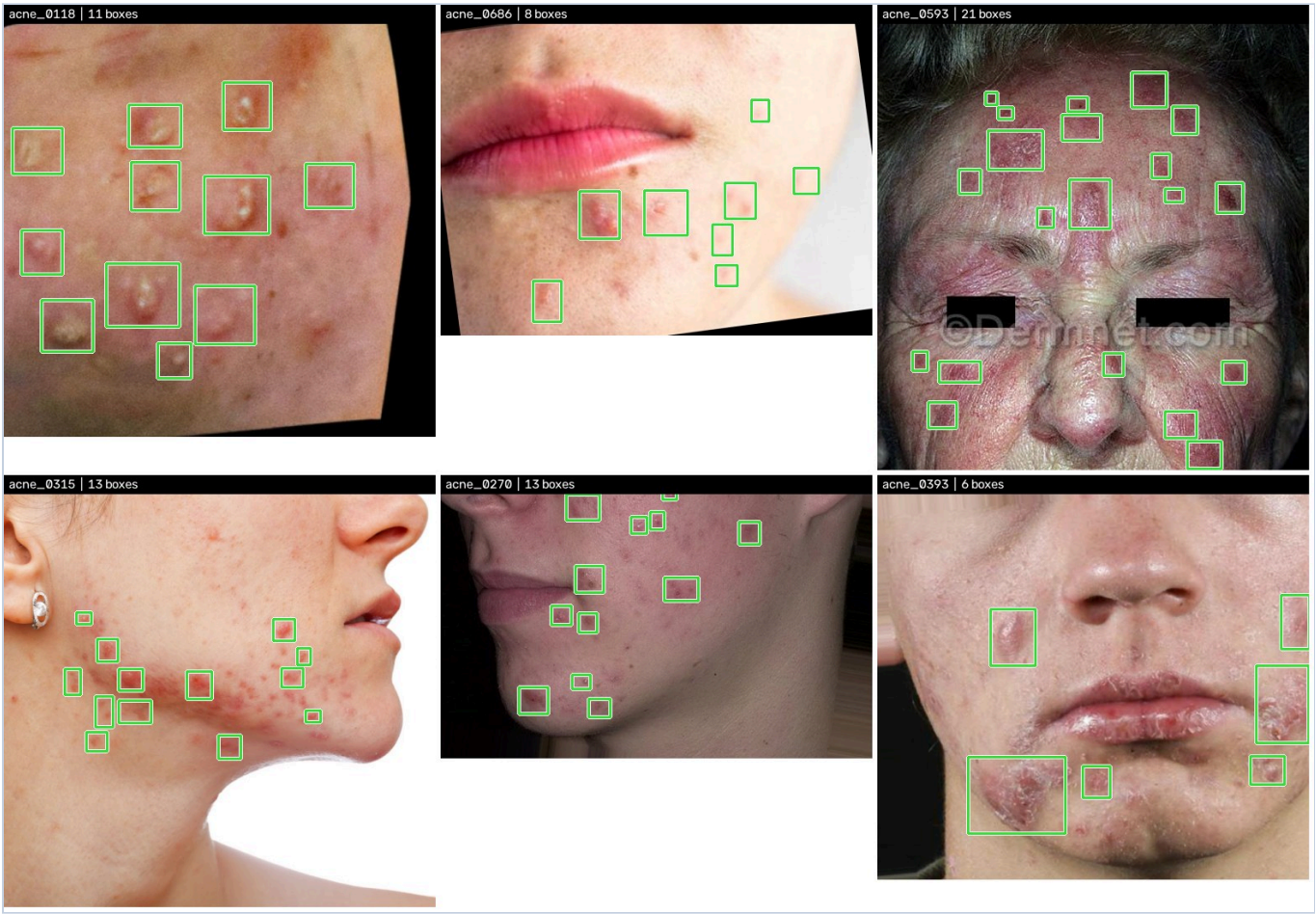
1. Look at the whole face at normal size first. Box only what you notice at that size as an area of skin darker than its surroundings. Do not zoom in to find more.
2. **One box = one connected patch.** If two dark areas touch, or the gap between them is smaller than the smaller one's width, they are one patch: one box around both.
3. Separate patches stay separate only if the gap is at least as wide as the smaller patch and normal skin is clearly visible between them.
4. Minimum size: do not box a mark smaller than about 7 % of the photo's short side.
5. Speckled areas (many tiny dots): one box around the area if it is clearly darker as a whole, otherwise skip it.
6. A patch stays inside one facial region (forehead, cheek, nose, upper lip, chin, temple); end the box at the region border.
7. Box edge: where the darkening has faded to about half. Do not include the soft rim and do not cut into the dark core.
8. Count check: typically 1 to 5 boxes; at 8 stop and merge touching neighbours.
9. No moles, scars, shadows, hair, beard or lip colour. If nothing qualifies, press X.
10. Do not look at older boxes or model drafts while boxing.

4. Acne: every boxing session

4.1 The photo pool and the first sprint (6 to 9 Oct 2026)

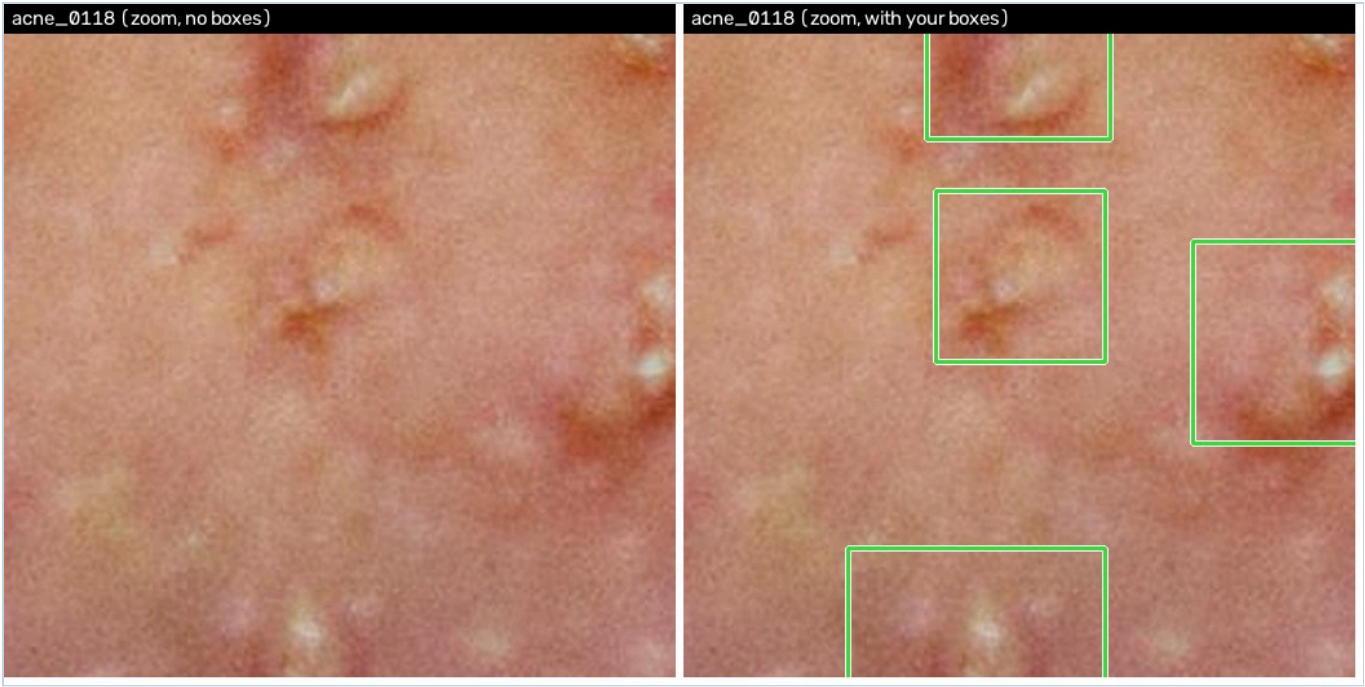
- Photo pool: **4,073 distinct acne photos** from your folder, de-duplicated (correlation 0.93), test photos reserved, listed in `sprint_photos.csv` and `sprint_assignments.csv`. Many are augmented copies from the public datasets (flipped, rotated, stretched borders), which is why some pictures below have streaked edges.
- You boxed them in the tool from scratch, starting with the facial photos (`acne_0000` onwards); the non-facial ones (`acne_nf_*`) came after.
- Result in `Elaiza__acne`: **927 photos reviewed: 658 boxed, 269 marked X**. Log: 980 actions, **6.2 hours**, 6 Oct 19:33 to 9 Oct 19:11.

Figure 1. Photos you boxed from scratch in the acne sprint (green = your boxes).



From left to right, top row: acne_0118 (11 boxes), acne_0686 (8), acne_0593 (21); bottom row: acne_0315 (13), acne_0270 (13), acne_0393 (6). One tight box per spot; touching spots get their own boxes.

Figure 2. Why tiny spots are hard: a zoom into acne_0118, without and with your boxes.

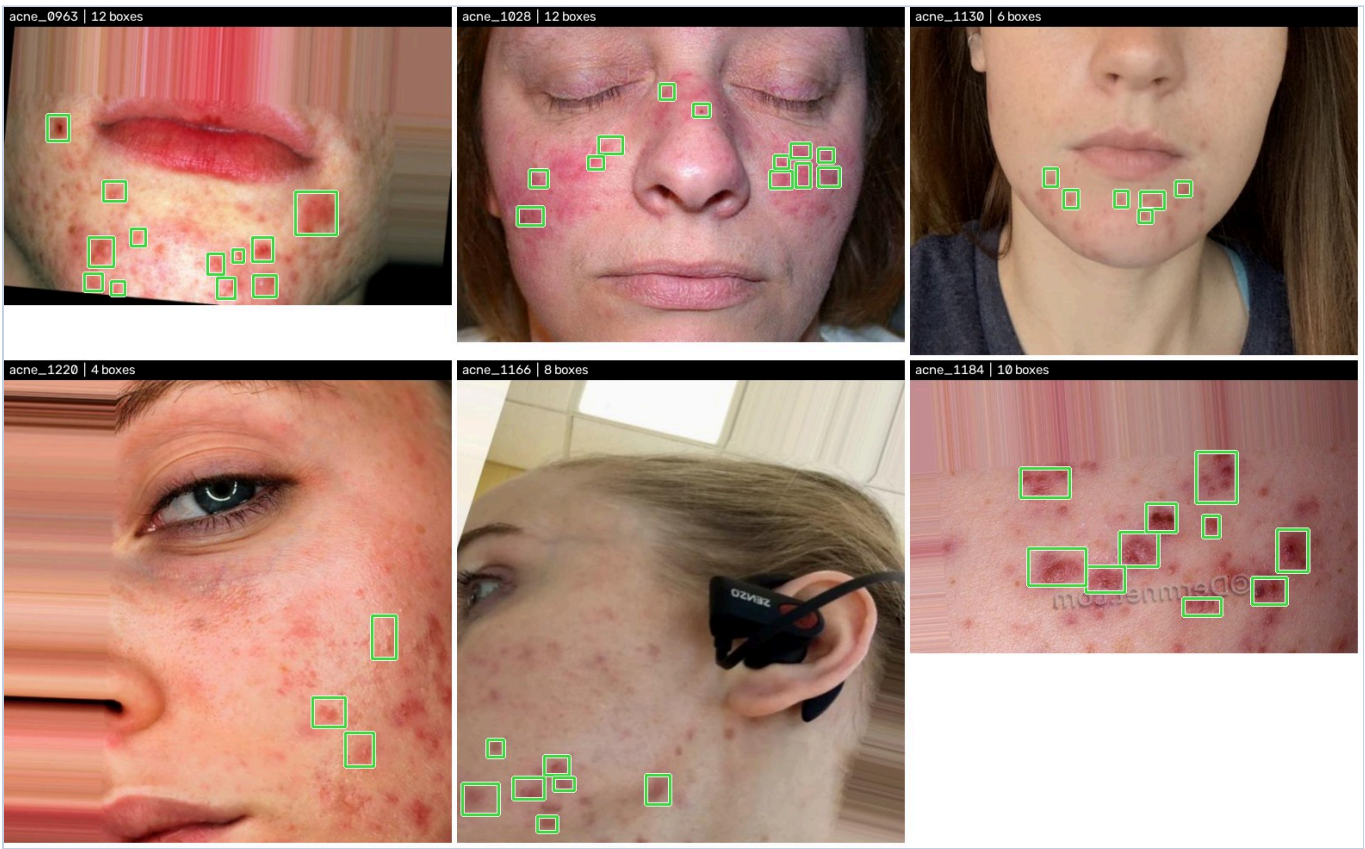


The spots are only a few dozen pixels wide, so a shift of a few pixels changes how well two boxes overlap.

4.2 The hand-boxed acne test photos (9 Oct)

- `Test_Acne`: 78 photos reserved for testing, **48 boxed, 30 marked X, 338 boxes**, boxed from scratch, 0.4 hours.

Figure 3. Hand-boxed acne test photos (green = your boxes).



acne_0963 (12 boxes), acne_1028 (12), acne_1130 (6) / acne_1220 (4), acne_1166 (8), acne_1184 (10).

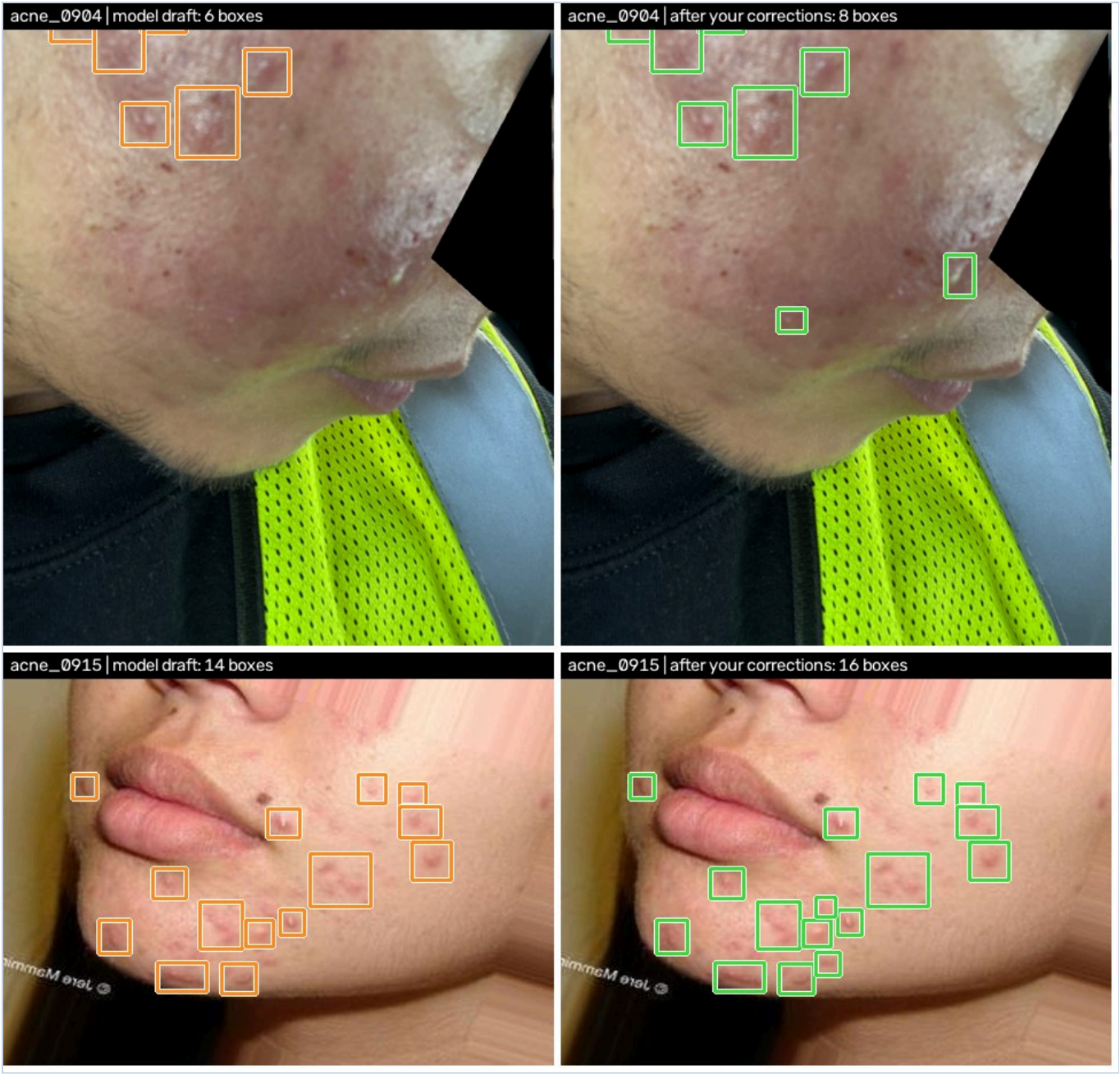
4.3 The helper drafts and how you corrected them

Because boxing 4,000 photos by hand is slow, a draft helper (a YOLOv8 model trained **only on your own boxes**, without the test photos) pre-drew boxes. You deleted wrong ones, added missed ones, and pressed **S**. Measured on your reviews:

Folder	Photos	Boxed	Marked X	Boxes	Tool time	Draft quality (measured on your corrections)
`Draft_Acne`	841	748	93	8,673	1.9 h	87 % of the final boxes were found by the draft (recall), 98 % of the draft boxes were kept (precision)
`More_Faces`	580	449	131	4,670	1.8 h	recall 79 %, precision 99.5 %

Measured speed: **14.2 seconds per photo** with drafts, faster than boxing from scratch. The drafts missed spots, which is why the corrections matter (the later checks found the training labels were still incomplete: see 4.4).

Figure 4. Model draft (orange) next to your corrected labels (green), same photo.



acne_0904: the draft had 6 boxes and the saved label has 8 after your corrections. acne_0915: 14 draft boxes, 16 after your corrections. The counts differ because you added spots the helper missed (and could delete wrong ones).

Important: a helper-drafted photo is **not** the same as a from-scratch photo. In the dataset manifest each photo records where it came from: `main` (618 photos, from scratch), `more` (448, More_Faces, drafts corrected), `drafts` (301, Draft_Acne, drafts corrected), `test_folder` (48), `fix46` (45), `fix6` (6). Total 1,466 photos, split into **1,145 training, 154 validation, 167 development test**. The validation set was boxed from scratch only; the test photos were never drafted by a model.

An important lesson about drafts: when 400 non-facial photos with helper drafts were reviewed and added (dataset v3), the test score **dropped** (mAP@0.5 0.387). Using only facial photos recovered it (0.423), so those drafted non-facial photos were removed. In the same way, the eczema drafts (made by the first detector) found only 36 % of the real boxes, so lesions the helper misses can end up unboxed and teach the detector the wrong thing.

4.4 Fixes you made after the checks

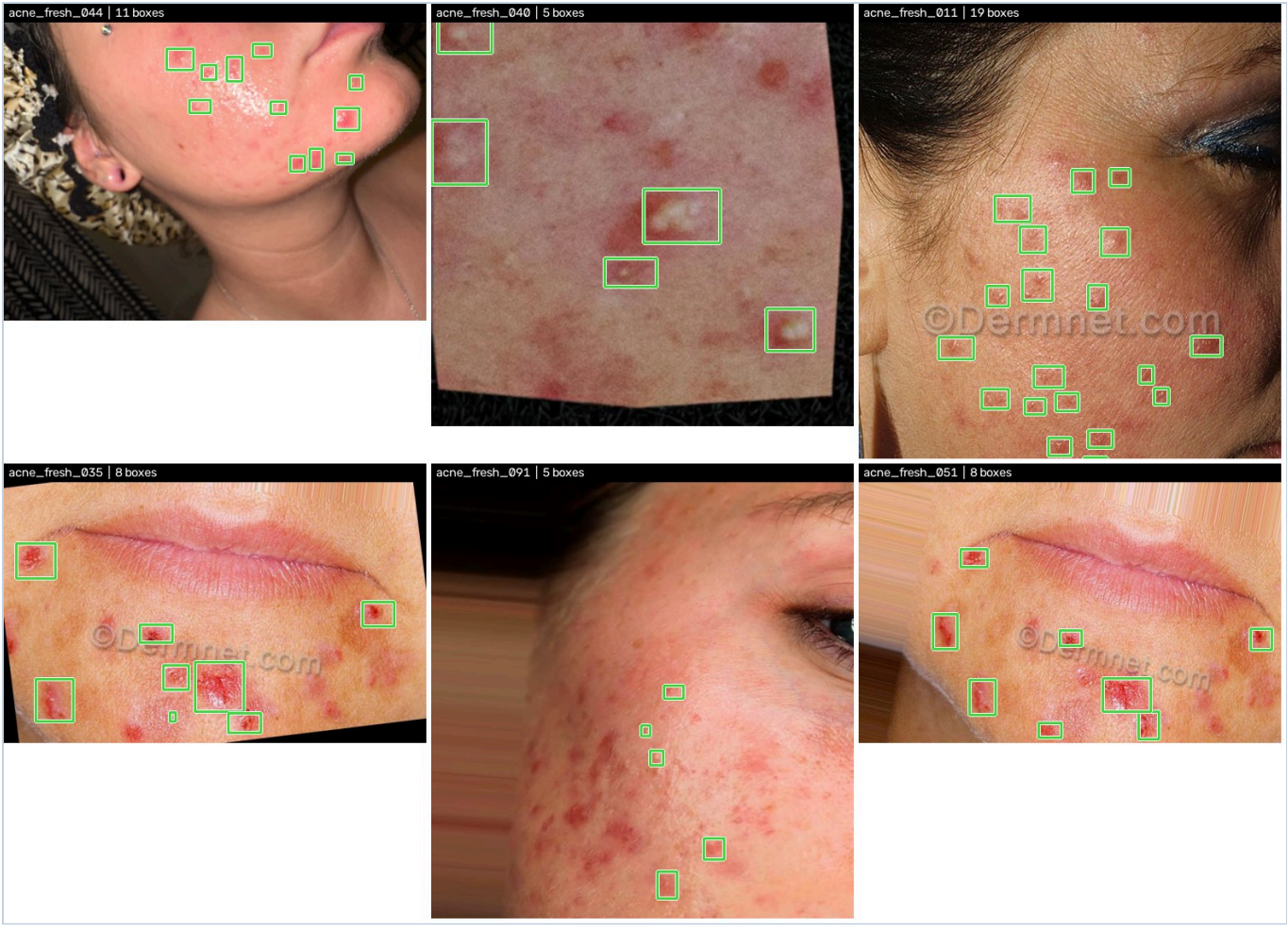
Folder	Why	Photos	Boxes after
`Fix_46_Doubled_Boxes`	the first check flagged photos with doubled or nested boxes. My rule was wrong for close-ups and touching neighbours: only 5 of the 46 were truly doubled. You fixed all 46 anyway.	46	673
`Fix_6_Test_Photos`	six test photos flagged	6	12
`Fix_Acne_Missed_Spots`	"second look": the trained model confidently saw spots you had not boxed. You kept the real ones and deleted the rest (60 photos).	60	**893 to 1,430 boxes (+60 %)**
`Acne_Review_Missing`	111 training photos where the model saw confident spots with no box near them (copies; your boxes preloaded)	111	2,078
`Acne_New_Photos`	250 new photos for experiments	250 (240 boxed)	2,907

The `Fix_Acne_Missed_Spots` result is the strongest sign that acne labels were incomplete: 60 photos grew from 893 to 1,430 boxes when re-reviewed with model help.

4.5 The independent and fresh test sets (no helper drafts)

- `Independent_Final_Set` (10 Oct): **43 photos, boxed blind and from scratch with no model run on them, 41 boxed, 2 X, 437 boxes**, 0.4 h. Scored once (single model mAP@0.5 0.358).
- `Acne_Fresh_Test` (15 Oct): **100 photos shuffled, renamed and blank, boxed from scratch: 36 with acne (203 boxes) and 64 marked X** (37 from the acne pool, 27 from the clear-skin pool), 0.1 h. Checked against every other photo for near-duplicates (correlation >= 0.90, 8 flips and rotations). Scored once for the installed 3-seed model: mAP@0.5 **0.336**.

Figure 5. Photos from the fresh acne test set, boxed from scratch (green).

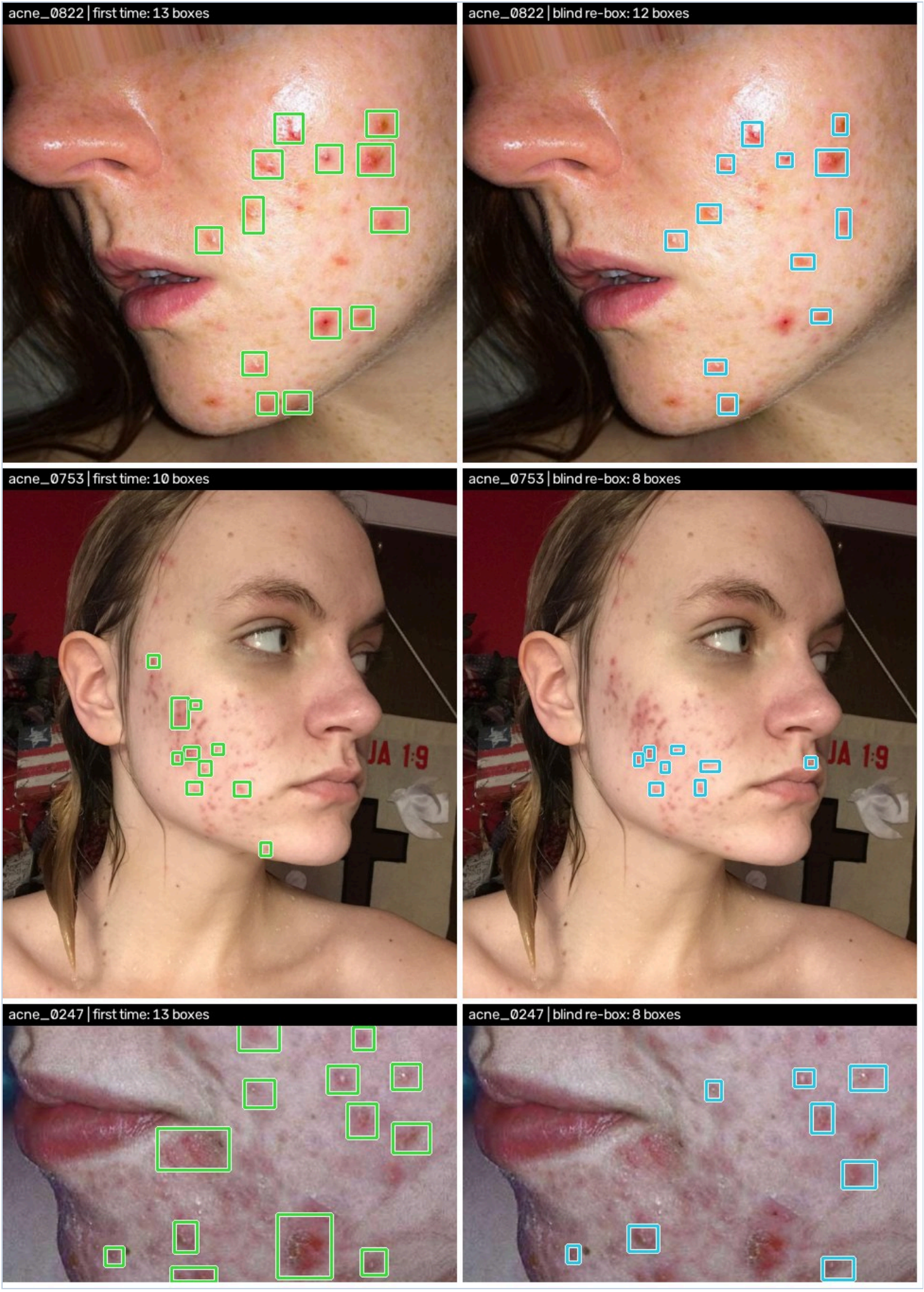


acne_fresh_044 (11), 040 (5), 011 (19), 035 (8), 091 (5), 051 (8). These photos were never used for training, validation, tuning or choosing a threshold.

4.6 How well do you agree with yourself? (acne)

`Reliability_Rebox_Acne` (13 Oct): **30 validation photos copied without their boxes, shuffled and renamed, boxed again blind (304 boxes, 0.3 h).** Compared with your first boxes: **F1 0.30 at IoU 0.5** (the score a second annotator would get against the first). That is the ceiling for strict box-level scores on tiny spots.

Figure 6. The same photo boxed twice by you, days apart (green = first time on 6 to 9 Oct, cyan = blind re-box on 13 Oct).


acne_0822 | first time: 13 boxes
acne_0822 | blind re-box: 12 boxes
acne_0753 | first time: 10 boxes
acne_0753 | blind re-box: 8 boxes
acne_0247 | first time: 13 boxes
acne_0247 | blind re-box: 8 boxes

acne_0822: 13 vs 12 boxes. acne_0753: 10 vs 8. acne_0247: 13 vs 8. The spots are mostly the same, but the exact edges, the very small spots and the ones in dense areas differ, which is why the overlap score between the two passes is low.

Further checks (`acne_box_scale_check.py` , `acne_box_margin_check.py`) showed the disagreement is mostly about where the edges go on tiny spots, not one consistent size convention.

4.7 Acne boxing in numbers

Folder	Purpose	Photos	Boxed / X	Boxes	Median box side	Tool time
`Elaiza__acne`	first sprint, from scratch	927 reviewed	658 / 269	not summed here	n/a	6.2 h
`Draft_Acne`	corrected helper drafts	841	748 / 93	8,673	0.067	1.9 h
`More_Faces`	corrected helper drafts	580	449 / 131	4,670	0.068	1.8 h
`Test_Acne`	hand-boxed test	78	48 / 30	338	0.055	0.4 h
`Independent_Final_Set`	blind independent set	43	41 / 2	437	0.058	0.4 h
`Acne_Fresh_Test`	fresh untouched test	100	36 / 64	203	0.053	0.1 h
`Reliability_Rebox_Acne`	agreement check	30	30 / 0	304	0.042	0.3 h
`Fix_Acne_Missed_Spots` , `Fix_46` , `Fix_6` , `Acne_Review_Missing` , `Acne_New_Photos`	fixes and reviews	473	all boxed	7,100	0.05 to 0.09	1.1 h
Total (listed folders)						**about 12.2 h**

The installed acne detector was trained on dataset v4 (1,145 photos, 11,621 boxes), which comes from the `main` , `more` , `drafts` , `fix46` and `fix6` sources above.

5. Hyperpigmentation: every boxing session

5.1 The first round (old rule)

- You boxed hyperpigmentation photos with the first rule: **646 photos opened = 504 boxed (2,117 boxes) + 142 marked X**. 100 of the hand-boxed photos were held out as the test set.
- A hyperpigmentation helper (`hyperpig_predraft.py` , trained on your boxes only, without the 100 test photos) drafted boxes on 600 photos; you reviewed them in `Draft_Hyperpig` : **583 boxed, 17 X, 3,247 boxes, 2.4 h**. Later checks showed that a second round of model-drafted boxes on the same photos did not improve the detector.
- Fixes: `Fix_Hyperpig_Boxes` (26 photos, 232 boxes) and `Fix_Hyperpig_Consistency` (42 photos, 487 boxes), 0.1 h each.

5.2 The discovery: the first rule was not reproducible

`Reliability_Rebox` (12 Oct): 30 validation photos boxed again blind: the re-box had **193 boxes (6.4 per photo) against 3.0 per photo in the original; agreement F1 0.28 at IoU 0.5**. 140 of the 193 new boxes had no match in the original; the median new box size was 0.09 against 0.125 for the original boxes. In other words you drew very different things on different passes, because "one box per patch" did not say what a patch is. This is the main reason the old scores were low (the old installed model scored mAP@0.5 0.18).

Other diagnosis findings that mattered: the public photos' boxes were about twice as coarse as yours and made up 75 % of training; the old validation set was in the public style; most training photos had near-duplicates; and your own boxing style drifted by about 50 % within one session.

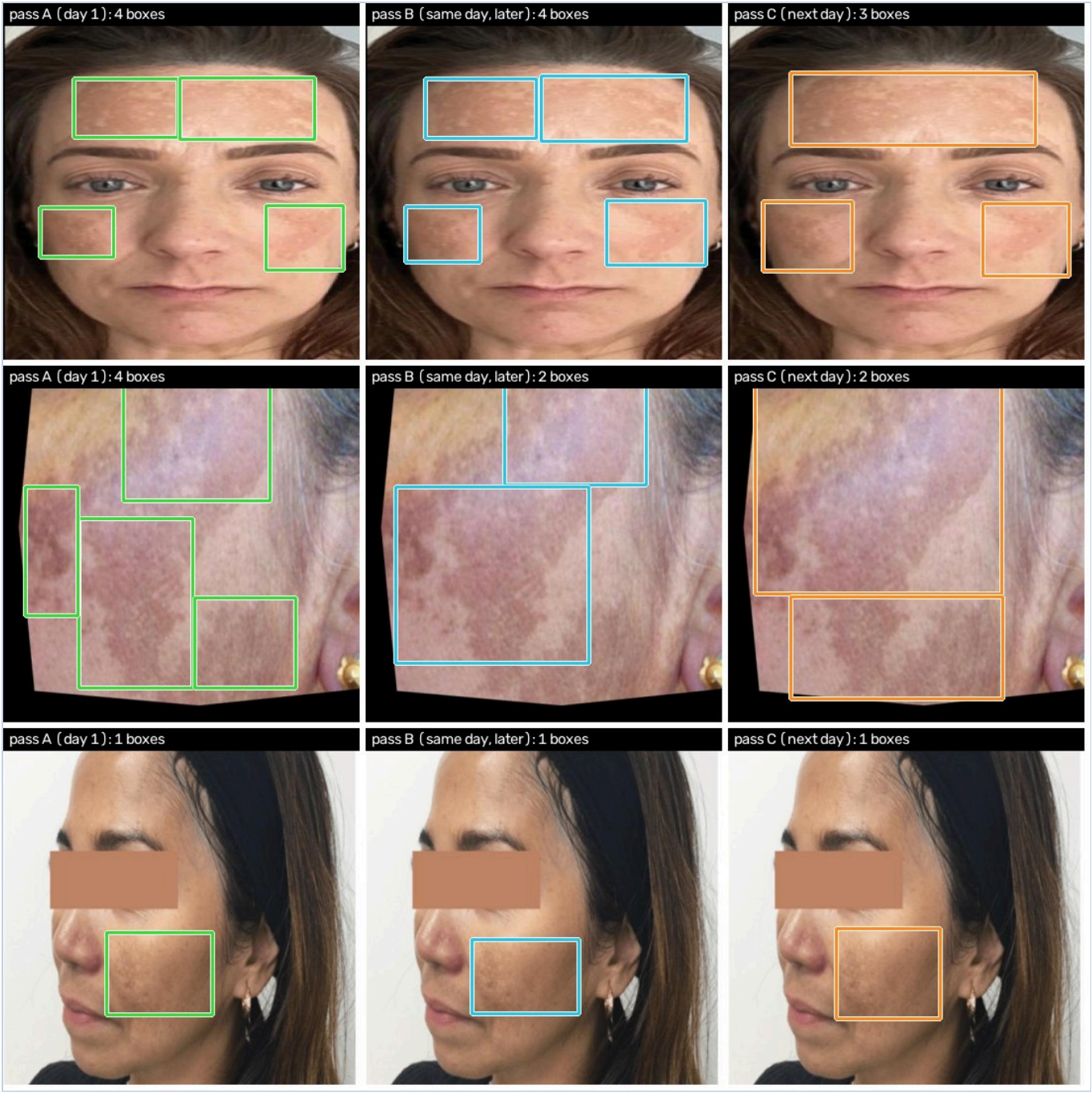
5.3 Rule v2 and the reproducibility test (12 to 13 Oct)

After writing rule v2 (section 3), you boxed the same 30 photos three times, blind, shuffled and renamed:

Pass	When	Boxes	Median box side	Tool time
A	12 Oct	86	0.22	0.1 h
B	12 Oct, a few minutes later	76	0.25	0.1 h
C	13 Oct (next day)	58	0.32	0.1 h

Agreement under rule v2: **F1 0.67 between A and B** (minutes apart, inflated by memory) and **0.49 to 0.55 at a day's gap**, against **0.28 with the first rule**. The number of boxes dropped and the boxes grew from pass to pass, which shows the rule still lets the box size drift upward, an honest limit.

Figure 7. The same photo boxed in pass A (green), pass B (cyan) and pass C (orange) under rule v2.



hand_hyperpigmentation_0220: 4 / 4 / 3 boxes. _0201: 4 / 2 / 2 (the largest patches are merged into fewer boxes as you apply "touching = one patch"). _0446: 1 / 1 / 1.

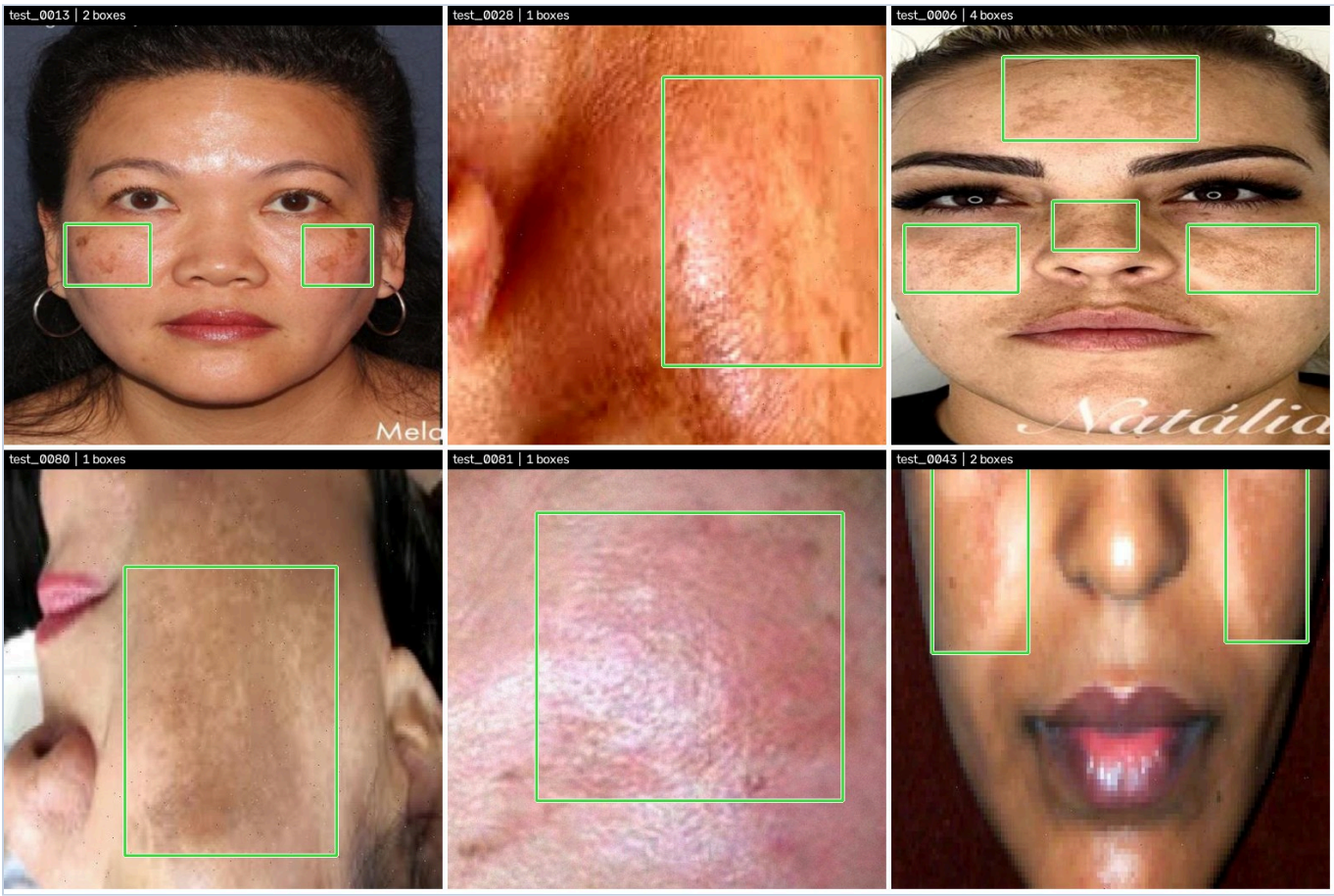
5.4 Relabelling everything under rule v2 (12 to 13 Oct)

Folder	What it is	Photos	Boxed / X	Boxes	Median side	Tool time
`V2_Test`	the new **test set**: 100 old test photos, blank, shuffled, renamed, boxed blind from scratch	100	98 / 2	226	0.255	0.6 h
`V2_Val`	validation (30 photos; with pass A these make the **60-photo validation set**)	30	27 / 3	56	0.28	0.2 h

Folder	What it is	Photos	Boxed / X	Boxes	Median side	Tool time
`V2_Train`	**808 training photos**: the old boxes converted to rule v2 by code, then you corrected each photo	808	794 / 14	1,589	0.262	1.6 h
`Negatives_100`	100 photos for measuring false alarms on clear skin (you marked X on most)	100	4 / 96	7	0.05	0.1 h

The code conversion (`hyp_v2_transform.py`) was fit on pass A and checked on pass B, so the check is not fitted to itself: boxes that touch or are closer than 1.0 times the smaller box's short side become one box (their union); boxes whose side is below 0.09 of the photo are dropped. The conversion alone gave a large gain; the first 200 hand-corrected photos gave most of the extra gain; correcting the other 600 added nothing measurable.

Figure 8. Hyperpigmentation test photos boxed from scratch under rule v2 (green).



test_0013 (2 boxes), test_0028 (1), test_0006 (4); test_0080 (1), test_0081 (1), test_0043 (2). One box per connected patch, edge where the darkening fades to half, no moles or scars boxed.

5.5 What the relabelling did to the scores (hyperpigmentation)

Stage	What changed	Test mAP@0.5
Old installed model (first rule, public + own boxes)	before this work	0.18 (on the old-rule test)

Stage	What changed	Test mAP@0.5
H20, boxes converted to rule v2 by code	rule v2 labels for validation and test, converted training boxes	0.341 (on the new rule-v2 test)
H23 to H24, 808 photos corrected by hand	your corrections of the first 200 photos	+0.09 on validation vs H20
H25, + 1,260 de-duplicated public photos, 3 seeds	final model	**0.508** (F1 0.520, 100 rule-v2 test photos, scored once)

Scores under rule v2 are **not comparable** with scores under the first rule: the box definition changed (boxes about twice as large).

6. What the boxes did to the detector results (summary)

	Acne	Hyperpigmentation
Your agreement with yourself (F1 at IoU 0.5)	**0.30** (tiny spots, edges differ by a few pixels)	**0.28 with the first rule**, **0.49 to 0.55 with rule v2**
What the labels were fixed with	missed-spot second look (+60 % boxes), doubled-box fix, a facial-only dataset	rule v2, blind relabelling of validation and test, hand-corrected training photos
Installed detector	3 seeds, YOLOv8s, 800 px, threshold 0.275	3 seeds (H25), YOLOv8s, 800 px, threshold 0.225
Cleanest test	fresh test (100 photos): mAP@0.5 **0.336**, F1 0.431	100 rule-v2 photos, scored once: mAP@0.5 **0.508**, F1 0.520
Other test	development test (167 photos): mAP@0.5 0.439, F1 0.507	3x3 face-zone F1 0.608
False alarms on photos without the condition	39 % of the 64 no-acne photos (15 % on clear skin)	about 32 % on clear skin

The main message of the boxing work: **the quality and reproducibility of the boxes limits the scores more than the model does.** For hyperpigmentation, a clear written rule plus careful relabelling raised mAP@0.5 from 0.18 to 0.51. For acne, the spots are too small for any rule to give a perfect agreement, so strict-overlap scores are capped near the level of your own agreement.

7. Scripts and tools behind the boxing

Script	What it does
`shared/lesion_box_tool.py`	the box editor
`shared/box_progress.py`	shows how far each annotator is
`acne/scripts/acne_predraft.py` , `hyperpigmentation/scripts/hyperpig_predraft.py`	train the draft helper on your boxes and pre-draw boxes
`acne/scripts/draft_more_faces.py` , `make_more_faces.py`	build the More_Faces folder and draft it
`acne/scripts/merge_acne_dataset.py`	builds the YOLO dataset (v1 to v4) only from photos you reviewed; leak checks; skips unreviewed machine drafts
`acne/scripts/acne_annotation_audit.py`	read-only audit: box sizes, doubled boxes, possible missing lesions
`acne/scripts/acne_second_look.py`	copies training photos with the model's confident extra boxes for you to accept or delete
`acne/scripts/make_acne_reliability.py` , `acne_reliability.py`	the 30-photo blind re-box and the agreement score

Script	What it does
`acne/scripts/make_independent_set.py`, `make_acne_fresh_test.py`	the independent and fresh test sets (shuffled, renamed, blank)
`hyperpigmentation/scripts/make_hyp_reliability_v2.py`, `hyp_reliability_v2.py`	the three blind passes and their agreement
`hyperpigmentation/scripts/hyp_v2_transform.py`	converts old boxes to rule v2 by code
`hyperpigmentation/scripts/make_hyp_v2_relabel.py`, `make_more_boxing.py`	review folders for relabelling
`hyperpigmentation/scripts/hyp_v2_train_consistency.py`	drift and correction checks of the 808 corrected photos
`shared/lesion_exp.py`	controlled experiments with a registry (one change per run, validation only)
`shared/negatives_eval.py`	false alarms on clear skin

8. Mistakes we made during the boxing work (so the record is honest)

- The "doubled or nested box" rule I wrote flagged touching neighbours and close-ups wrongly: only 5 of 46 flagged acne photos were truly doubled. The rule was loosened.
- The plan to draft 400 new facial photos was wrong because the facial pool was already used up; the non-facial drafts that were reviewed hurt the test score and were removed.
- The first hyperpigmentation rule was not reproducible (agreement 0.28). The score gain from rule v2 depended on writing a clearer rule, not on changing the model.
- Boxes still drift larger from pass A to pass C under rule v2 (median side 0.22, 0.25, 0.32), so the rule is better but not perfect.
- The acne development test set influenced earlier decisions and was scored several times, so a fresh set was built; the fresh result (0.336) is lower than the development result (0.439).
- Early machine-drafted boxes in `claude_boxed_Acne` were never reviewed and were not used.
- Some of the public photos carry watermarks, are augmented copies, or show children or other people; they are public-dataset photos, not consented participant photos. The pool may also contain photos of children.

9. Limits to state

One annotator; same-annotator agreement is low for tiny acne spots and moderate for hyperpigmentation; rule-v2 boxes are whole patches about twice the size of the first rule's; tool-logged hours count active time only; public photos are not consented kiosk images; no dermatologist has verified the boxes.